CIA-1

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1. a)

1. Data Link Layer:

The Data Link Layer is the second layer of the OSI model. It provides reliable communication between two directly connected devices.

Functions:

1. Framing: Divides the received data into small units called frames.
2. Physical Addressing: Uses MAC addresses to identify the source and destination devices.
3. Error Detection: Detects transmission errors using CRC.
4. Flow Control: Controls the speed of data transmission between directly connected devices.
5. Media Access Control: Determines which device can use the communication medium.
6. Error Control: Detects damaged or lost frames and may support retransmission.

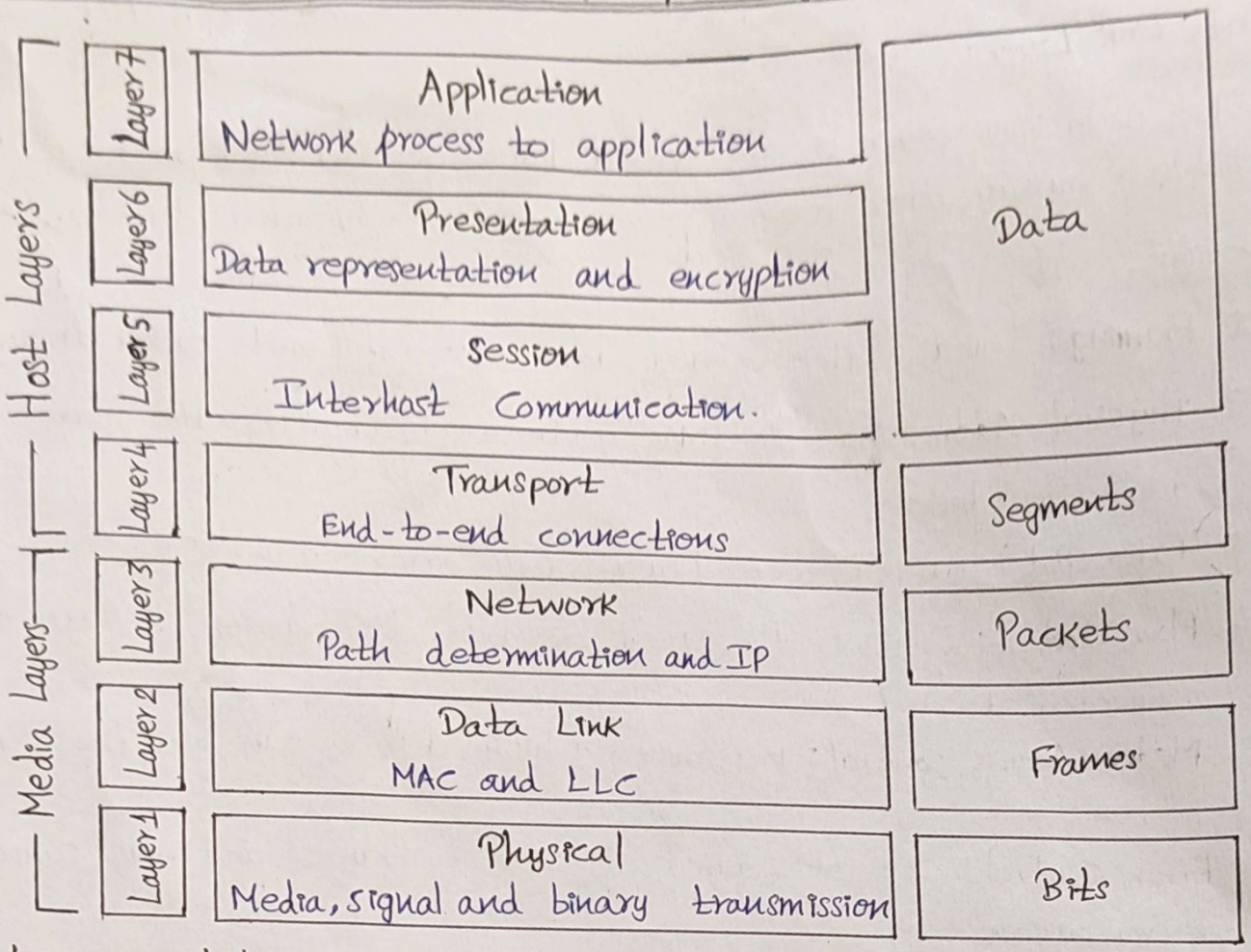
2. Transport Layer:

The Transport Layer is the fourth layer of the OSI model. It provides end-to-end communication between applications running on different devices.

Functions:

1. Segmentation: Divides large messages into smaller segments.
2. Port Addressing: Uses port numbers to identify the sending and receiving applications.
3. Flow Control: Prevents the sender from overwhelming the receiver.
4. Error Control: Uses acknowledgements and retransmissions to recover lost segments in TCP.
5. Congestion Control: Controls network traffic to prevent congestion.
6. Reassembly: Combines received segments into the original message.

Diagram: Working of Data Link and Transport Layers



Layered representation:

SENDER – Application Data – Transport Layer – Data Link Layer – Physical L
RECEIVER – Transport Layer – Data Link Layer – Physical Layer – TRANSMISSION

How both layers work together:

* The Data Link Layer provides node-to-node delivery over an individual l
* The Transport Layer provides end-to-end delivery between applicati
* The Data Link Layer checks individual frames for errors.
* The Transport Layer uses sequence numbers, acknowledgements and retransmissions to provide reliable delivery through TCP.

Conclusion:

Both layers work together to ensure efficient, accurate and reliable data communication.

b) TCP vs UDP

Feature	TCP	UDP
Connection	Connection-Oriented: establishes connection first	Connectionless: no setup
Reliability	Reliability delivers with ACKs and retransmission	Best-effort delivery
Speed	Generally slowers due to reliability mechanisms	Generally faster, with low overhead.
Header Size	Minimum 20 bytes	8 bytes
State	Maintains connection state	No connection state required
Examples	HTTP/HTTPS, SMTP, FTP	DNS queries, VOIP, online gaming, streaming

TCP is preferred when reliable, ordered delivery is required, UDP is preferred when low delay and low overhead are important.

2. a) Encapsulation and Decapsulation:

Encapsulation is the process in which each layer at the sender adds its own control information to the data received from the layer above.

Steps:

1. Application layer creates data
2. Transport Layer adds a header, forming a segment (TCP or datagram (UDP).

3. Network Layer adds a header and trailer, forming a frame.
4. Data link layer adds IP header, forming a packet.
5. Physical layer transmits the frame as bits.

Decapsulation occurs at the destination. Each layer removes its corresponding header (and the Data Link trailer) and passes the remaining data upward until it reaches the application.

Diagram:

Steps:

1. Physical Layer: Receives bits from the transmission medium.
2. Data Link Layer: Reconstructs the frame, checks for error and removes the header and trailer.
3. Network Layer: Removes the IP address and passes the packet payload upward.
4. Transport Layer: Removes the TCP/UDP header and reassembles the segments.
5. Application Layer:- Receives the original data and delivers is to the application.

Diagram:

Encapsulation at Source

Application Layer

Role of Intermediate Routes:

1. The Route receives the frame at its incoming interface.
2. The Data Link Layer removes the frame header & trailer.
3. The Network Layer examines the destination IP address.
4. The Router selects the next hop using its routing Table.
5. It encapsulates the same IP packet into a New Data Link frame for the next link.
6. The new frame is transmitted towards the destination.

b) Functions of the Network Layer:

1. Logical addressing:

User source and destination IP address to identify hosts and networks.

2. Routing:

Routers use routing tables and the destination I/P address to select the best path.

3. IP vs MAC address:

IP address identify endpoints across networks and normally remain the same end to end. MAC addresses are used for delivery on a local link and normally change each routed hop.

4. Fragmentation:

Divides an IP packet into smaller frames when permitted and required by the network MTU.

Time to Live (TTL) :

* TTL is a field in the IPV4 header.
* Each Router decreases into values by at least one.
* When TTL reaches zero, the packet is discarded to prevent infinite routing loops.

IP Address (Logical)	MAC address (Physical)
Works at Network Layer	Works at Data Link Layer.
Identifies source and destination across networks.	Identifies device on a local link.
Uses for routing packets	Used for frame delivery within a link.
Normally remains the same end-to-end.	Changes at each routed hop.
Eg: 192.168.1.10	Eg: 00:1A:2B:3C:4D:5E

Packet Routing Through Routers

Source -> Router 1 -> Router 2 -> Destination.

3. a)

1. Classful IP addressing

-> Classful IP addressing divides IPV4 addresses into five classes (A, B, C, D) and E). Classes A, B & C are commonly used for unicast networks.

Class	Default Subnet Mask	Network / Host
A	255.0.0.0 (/8)	8 networks bits, 24 host bit
B	255.255.0.0 (/16)	16 network bits, 16 host bits.
C	255.255.255.0 (/24)	24 network bits, 8 host bits.

Limitations:

1. Fixed Network Boundaries
2. Wastage of IP addresses.
3. Rapid IPv4 Exhaustion.
4. Inflexible Subnetting.
5. Large Routing Tables.
6. CIDR (classless Inter - Domain Routing)

CIDR (classless Inter - Domain Routing):

CIDR is a method of allocating IP addresses using variable-length network prefixes instead of fixed classes.

* It uses slash notations, such as /22 or /26.
* It supports variable length subnet Masking (VLSM)
* It allocates IP addresses according to organizational requirements.
* It supports route aggregation, reducing routing table entries.

b) Subnetting Calculation:

Given:

* IP address : 192.168.25.130
* Subnet Mask : 255.255.255.192
* CIDR Notation : /26

STEP 1: Convert the last octet into binary.

IP address last octet:

130 = 10000010

Subnet Mask Last octet:

192 = 11000000

Perform the AND operation:

IP address : 10000010

Subnet Mask : 11000000

Network ID : 10000000 = 128

∴ The Network ID is 192.168.25.128 /26.

STEP 2: Calculate Block Size:

Block size = 256 - 192 = 64

STEP 3: Calculate the broadcast address:

The block start at .128 and ends at .191

Broad Cast address = 192.168.25.191

STEP 4: Calculate usuable host address:

Host bits = 32 - 26 = 6

Usable hosts = $2^6 - 2$ = 62

Parameter	Answer
Network IP	192.168.25.128
First usable IP	192.168.25.129
Last usable IP	192.168.25.190
Broadcast address	192.168.25.191
Block Size	64
Usable hosts	62

4. Subnetting of 172.16.0.0/16

a) Minimum Borrowed Bits and Subnet Mask

Given network:

172.16.0.0/16

Required:

At least 6 subnets

$2^n \geq 6$

$2^3 = 8$ subnets

Borrowed bits = 3

New prefix = /16 + 3 = /19

* Subnet Mask: 255.255.224.0
* CIDR: /19

b) Number of Subnets and Hosts:

Host bits = 32 - 19 = 13

Total Subnets:

$2^3 = 8$

Usuable host per subnet:

$2^{13} - 2 = 8190$

* Total Subnets = 8
* Usuable hosts per subnet = 8190
* Unused subnets = 2, if only 6 are required.

c) First Two Subnets:

Block Size = 256 - 224 = 32

Details	First Subnet	Second Subnet
Network ID	172.16.0.0	172.16.32.0
First usable IP	172.16.0.1	172.16.32.1
Last usable IP	172.16.31.254	172.16.63.254
Broadcast	172.16.31.255	172.16.63.255.

5. a) Static Routing Vs Dynamic Routing:

Feature	Static Routing	Dynamic Routing
Configuration	Manual	Automatic
Topology changes	Manual updates needed	Automatically adapts.
Convergence	No routing - protocol convergence	Requires Convergence time.
Scalability	Small network	Large Network.
CPU & Memory	Low Usuage	Higher Usage
Bandwidth	No routing updates	Uses bandwidth for updates.
Security	No routing updates to Spoof	Requires protocol security
Administrative distance	1 (default)	RIP:120. DSPF:110

Dynamic routing protocols:

Uses of static routing:

* Small networks
* Sub networks with the exist path.
* Default routes to an ISP.
* When low Overhead and predictable routing are required.

b) Static Route Configuration:

Requirements:

Route 192.168.2.0/24 through next-hop 10.0.0.2.

STEP 1: Enter privileged mode.

Router > enable

STEP 2: Enter configuration mode

Router # configure terminal

STEP 3: Configure the static route.

Router (config) # ip route 192.168.2.0 255.255.255.0 10.0.0

STEP 4: Exist Configuration mode.

Router (config) # exist

STEP 5: Save Configuration.

Router # Copy running-config startup-config.

STEP 6: Verify the route

Router # show ip route

STEP 7: Test Connectivity

Router # ping 192.168.2.1

The router forwards packets for 192.168.2.0/24 through 10.0.0.2.